

Toy DMP Model in CT

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June 20, 2026

This note follows a lot from the note by Toshihiko Mukoyama.

1 Unemployment Dynamics

Let the period length be Δ and unemp. rate at time t be $u(t)$:

$$u(t + \Delta) = \sigma \Delta (1 - u(t)) + (1 - \lambda_w \Delta) u(t)$$

where $\sigma \Delta > 0$ is the prob. of losing the job, $\lambda_w \Delta > 0$ be the prob. of finding a job over Δ .

We can rewrite it as:

$$\begin{aligned} u(t + \Delta) &= \sigma \Delta - \sigma \Delta u(t) + u(t) - \lambda_w \Delta u(t) \\ u(t + \Delta) - u(t) &= \sigma \Delta - \sigma \Delta u(t) - \lambda_w \Delta u(t) \\ \frac{u(t + \Delta) - u(t)}{\Delta} &= \sigma - \sigma u(t) - \lambda_w u(t) \\ \frac{u(t + \Delta) - u(t)}{\Delta} &= \sigma (1 - u(t)) - \lambda_w u(t) \end{aligned}$$

so taking $\Delta \rightarrow 0$:

$$u'(t) = \sigma (1 - u(t)) - \lambda_w u(t)$$

We then try to endogenize job-finding rate λ_w . Let agg. vacancy be v and the matching function takes the form of

$$M(u, v) \Delta$$

Some assumptions:

1. matching is random
 - (a) all vacancies have an equal chance to match w/ an employed worker
 - (b) all unemployed workers have an equal chance to match w/ a vacancy
2. M is increasing in both terms
3. M is CRS

Under these assumptions, the prob. of a vacancy matching w/ an unemp. worker during Δ is

$$\frac{M(u, v) \Delta}{v} = M\left(\frac{1}{\theta}, 1\right) \Delta = \lambda_f(\theta) \Delta$$

where $\theta \equiv \frac{v}{u}$ is the tightness. The prob. of an unemp. worker matching w/ a vacancy during Δ is:

$$\frac{M(u, v) \Delta}{u} = M(1, \theta) \Delta = \lambda_w(\theta) \Delta$$

and $\lambda_w(\theta) = M(1, \theta) = \theta M\left(\frac{1}{\theta}, 1\right) = \theta \lambda_f(\theta)$. So unemp. rate dynamics now become:

$$u'(t) = \sigma(1 - u(t)) - \lambda_w(\theta(t)) u(t)$$

1.1 Beveridge curve condition

Consider ss. where $v(t) \equiv v, \forall t$ and $u'(t) = 0$:

$$\begin{aligned} 0 &= \sigma(1 - u) - \lambda_w\left(\frac{v}{u}\right) u \\ \lambda_w\left(\frac{v}{u}\right) u &= \sigma(1 - u) \\ u\left(\sigma + \lambda_w\left(\frac{v}{u}\right)\right) &= \sigma \\ u &= \frac{\sigma}{\sigma + \lambda_w\left(\frac{v}{u}\right)} \end{aligned}$$

This $u = \frac{\sigma}{\sigma + \lambda_w\left(\frac{v}{u}\right)}$ is referred to as the Beveridge curve condition.

2 DMP Model

2.1 Workers and Firms

The DMP builds on the unemp. dynamics and adds the endo. determination of agg. vacancy $v(t)$. Firms create vacancies by trading off the current vacancy posting cost and future profit from the firm-worker match.

We assume that each vacancy hires one worker. Let $w(t)$ be the eq. wage. The PV of being employed, in discrete-time setting w/ period length Δ is

$$W(t) = w(t)\Delta + \frac{1}{1+r\Delta} ((1-\sigma\Delta)W(t+\Delta) + \sigma\Delta U(t+\Delta)) \quad (1)$$

where $U(t)$ is the value of being unemployed.

$$U(t) = b\Delta + \frac{1}{1+r\Delta} (\lambda_w(\theta(t))\Delta W(t+\Delta) + (1-\lambda_w(\theta(t))\Delta)U(t+\Delta))$$

Multiplying $1+r\Delta$ on both sides of (1) we get:

$$\begin{aligned} (1+r\Delta)W(t) &= (1+r\Delta)w(t)\Delta + ((1-\sigma\Delta)W(t+\Delta) + \sigma\Delta U(t+\Delta)) \\ r\Delta W(t) &= (1+r\Delta)w(t)\Delta + ((1-\sigma\Delta)W(t+\Delta) + \sigma\Delta U(t+\Delta)) - W(t) \\ rW(t) &= (1+r\Delta)w(t) + \sigma U(t+\Delta) + \left(\frac{1}{\Delta} - \sigma\right)W(t+\Delta) - \frac{1}{\Delta}W(t) \\ rW(t) &= (1+r\Delta)w(t) + \sigma(U(t+\Delta) - W(t+\Delta)) + \frac{W(t+\Delta) - W(t)}{\Delta} \end{aligned}$$

so taking $\Delta \rightarrow 0$:

$$rW(t) = w(t) + \sigma(U(t) - W(t)) + W'(t)$$

Similarly for $U(t)$:

$$\begin{aligned} (1+r\Delta)U(t) &= (1+r\Delta)b\Delta + (\lambda_w(\theta(t))\Delta W(t+\Delta) + (1-\lambda_w(\theta(t))\Delta)U(t+\Delta)) \\ r\Delta U(t) &= (1+r\Delta)b\Delta + (\lambda_w(\theta(t))\Delta W(t+\Delta) + (1-\lambda_w(\theta(t))\Delta)U(t+\Delta)) - U(t) \\ rU(t) &= (1+r\Delta)b + (\lambda_w(\theta(t))W(t+\Delta) - \lambda_w(\theta(t))U(t+\Delta)) + \frac{U(t+\Delta) - U(t)}{\Delta} \end{aligned}$$

so taking $\Delta \rightarrow 0$:

$$rU(t) = b + \lambda_w(\theta(t))(W(t) - U(t)) + U'(t)$$

Because of the one-firm to one-worker assumption, the firm is either matched or vacant, the value of a matched firm is:

$$J(t) = p(t)\Delta - w(t)\Delta + \frac{1}{1+r\Delta}(\sigma\Delta V(t+\Delta) + (1-\sigma\Delta)J(t+\Delta))$$

where $p(t)\Delta$ is the product produced by the match and $V(t)$ is the value of vacancy, computed by:

$$V(t) = -\kappa\Delta + \frac{1}{1+r\Delta}(\lambda_f(\theta(t))\Delta J(t+\Delta) + (1-\lambda_f(\theta(t))\Delta)V(t+\Delta))$$

where $\kappa\Delta$ is the cost of posting a vacancy. It's then easy to see that

$$rJ(t) = p(t) - w(t) + \sigma(V(t) - J(t)) + J'(t)$$

and

$$rV(t) = -\kappa + \lambda_f(\theta(t))(J(t) - V(t)) + V'(t) \quad (2)$$

We assume that anyone can start a firm by posting a vacancy

1. if $V(t) > 0$, new entrants would keep posting vacancies until $V(t)$ becomes 0. Additional vacancies would drop the value of $V(t)$ because $\lambda_f(\theta(t))$ is a decreasing function of $\theta(t) = \frac{v(t)}{u(t)}$. Therefore $V(t) = 0$ has to always hold, as long as $v(t) > 0$.¹

2. We call the condition

$$V(t) = 0$$

the *free-entry condition*

One can formalize 1 as follows:

Let $u(t)$ be the state var. at time t , which evolves as

$$u' = \sigma(1-u) - \lambda_w u$$

¹It's possible that $V(t) < 0$ when $v(t) = 0$.

and $v(t)$ be the control var. and $v \geq 0$. Given (2) and taking $u(t), J(t), V'(t)$ as given, we have

$$V(t) = \frac{\lambda_f(\theta(t)) J(t) - \kappa + V'(t)}{r + \lambda_f(\theta(t))}$$

so

$$\frac{\partial V}{\partial \lambda_f} = \frac{J(t)(r + \lambda_f(\theta(t))) - (\lambda_f(\theta(t)) J(t) - \kappa + V'(t))}{(r + \lambda_f(\theta(t)))^2}$$

at ss., $V'(t) = 0$, so we have

$$\begin{aligned} \frac{\partial V}{\partial \lambda_f} &= \frac{r J(t) + \kappa - V'(t)}{(r + \lambda_f(\theta(t)))^2} \\ &= \frac{r J(t) + \kappa}{(r + \lambda_f(\theta(t)))^2} > 0 \end{aligned}$$

Given the matching function, it's easy to see that $\lambda_f(\theta) = M(\frac{1}{\theta}, 1)$ is decreasing in θ :

$$\frac{\partial \lambda_f}{\partial v} = \lambda'_f(\theta) \frac{1}{u} < 0$$

so

$$\frac{\partial V}{\partial v} = \frac{\partial V}{\partial \lambda_f} \frac{\partial \lambda_f}{\partial v} < 0$$

Therefore, if $V(t) > 0$, then it follows that vacancy goes up and in turn pushing $V(t) \rightarrow 0$. If $V(t) < 0$, then nobody is willing to post a vacancy and in turn $v(t) = 0$. Finally if $V(t) = 0$, then entrants are indifferent and therefore $\forall v(t) \geq 0, s.t. V(v(t)) = 0$ will work.

2.2 Wage Determination

In this framework, for a worker to meet w/ another firm, the worker has to endure the cost of unemployment. Similarly, for the firm, losing a worker would imply waiting for another worker by incurring the vacancy cost. Thus, the matched worker and firm are in a bilateral monopoly situation.

Here we assume that the surplus is split by the level of wage that solves the following maximization problem:

$$\max_w (\tilde{W}(w, t) - U(t))^\gamma (\tilde{J}(w, t) - V(t))^{1-\gamma} \quad (3)$$

where $\gamma \in (0, 1)$. The functions $\tilde{W}(w, t)$ and $\tilde{J}(w, t)$ are the values of employed workers and a matched firm, here we make it explicit that only w in this period can move. The solution to this problem is often called *Generalized Nash Bargaining Rule*.

The functions $\tilde{W}(w, t)$ and $\tilde{J}(w, t)$ can be written as:

$$\tilde{W}(w, t) = w\Delta + \frac{1}{1+r\Delta} ((1-\sigma\Delta)W(t+\Delta) + \sigma\Delta U(t+\Delta))$$

and

$$\tilde{J}(w, t) = p(t)\Delta - w\Delta + \frac{1}{1+r\Delta} (\sigma\Delta V(t+\Delta) + (1-\sigma\Delta)J(t+\Delta)).$$

By taking FOC for (3), we have:

$$\begin{aligned} \max_w \gamma \ln(\tilde{W}(w, t) - U(t)) + (1-\gamma) \ln(\tilde{J}(w, t) - V(t)) \\ 0 = \frac{\gamma}{\tilde{W}(w, t) - U(t)} \frac{\partial \tilde{W}(w, t)}{\partial w} + \frac{1-\gamma}{\tilde{J}(w, t) - V(t)} \frac{\partial \tilde{J}(w, t)}{\partial w} \end{aligned}$$

Given $\frac{\partial \tilde{W}(w, t)}{\partial w} = \Delta$, $\frac{\partial \tilde{J}(w, t)}{\partial w} = -\Delta$, we have

$$\begin{aligned} 0 &= \frac{\gamma}{\tilde{W}(w, t) - U(t)} \Delta - \frac{1-\gamma}{\tilde{J}(w, t) - V(t)} \Delta \\ 0 &= \frac{\gamma}{\tilde{W}(w, t) - U(t)} - \frac{1-\gamma}{\tilde{J}(w, t) - V(t)} \\ \frac{1-\gamma}{\tilde{J}(w, t) - V(t)} &= \frac{\gamma}{\tilde{W}(w, t) - U(t)} \\ (1-\gamma)(\tilde{W}(w, t) - U(t)) &= \gamma(\tilde{J}(w, t) - V(t)) \end{aligned} \tag{4}$$

so the w^* that satisfies (4) is the eq. wage.

2.3 Steady State Equilibrium

First, consider the ss. eq. where $u(t)$ and $v(t)$ are constant. With the conditions above we have

$$rW = w - \sigma(W - U)$$

$$rU = b + \lambda_w(\theta)(W - U)$$

$$rJ = p - w - \sigma(J - V)$$

$$rV = -\kappa + \lambda_f(\theta)(J - V)$$

$$V = 0$$

$$(1 - \gamma)(W - U) = \gamma(J - V)$$

in total 6 equations w/ 6 unknowns W, U, J, V, θ, w . We can then try to reduce this system to 1 to 1. It can be shown that we have

$$(1 - \gamma)(p - b) - \frac{r + \sigma + \gamma\lambda_w(\bar{\theta})}{\lambda_f(\bar{\theta})}\kappa = 0.$$

This condition is then used to solve for θ and is often referred to as the *job creation condition*.

For simple math, we have

$$rV = -\kappa + \lambda_f(\theta)(J - V)$$

$$0 = -\kappa + \lambda_f(\theta)(J - 0)$$

$$J = \frac{\kappa}{\lambda_f(\theta)}$$

Given

$$rJ = p - w - \sigma(J - V)$$

$$rJ = p - w - \sigma J$$

$$J = \frac{p - w}{r + \sigma}$$

we have the free-entry condition on firm side:

$$\frac{\kappa}{\lambda_f(\theta)} = \frac{p-w}{r+\sigma}$$

Given

$$rW = w - \sigma(W - U)$$

$$rU = b + \lambda_w(\theta)(W - U)$$

$$r(W - U) = w - b - (\sigma + \lambda_w(\theta))(W - U)$$

$$(r + \sigma + \lambda_w(\theta))(W - U) = w - b$$

$$W - U = \frac{w - b}{r + \sigma + \lambda_w(\theta)}$$

so given $(1 - \gamma)(W - U) = \gamma(J - V)$, we have

$$(1 - \gamma) \frac{w - b}{r + \sigma + \lambda_w(\theta)} = \gamma \frac{\kappa}{\lambda_f(\theta)}$$

$$w - b = \frac{\gamma}{1 - \gamma} \frac{\kappa}{\lambda_f(\theta)} (r + \sigma + \lambda_w(\theta))$$

$$p - w = p - \left[\frac{\gamma}{1 - \gamma} \frac{\kappa}{\lambda_f(\theta)} (r + \sigma + \lambda_w(\theta)) + b \right]$$

$$(r + \sigma) \frac{\kappa}{\lambda_f(\theta)} = p - \left[\frac{\gamma}{1 - \gamma} \frac{\kappa}{\lambda_f(\theta)} (r + \sigma + \lambda_w(\theta)) + b \right]$$

$$(r + \sigma) \kappa = \lambda_f(\theta) (p - b) - \frac{\gamma \kappa}{1 - \gamma} (r + \sigma + \lambda_w(\theta))$$

$$(r + \sigma) \kappa + \frac{\gamma \kappa}{1 - \gamma} (r + \sigma + \lambda_w(\theta)) = \lambda_f(\theta) (p - b)$$

$$(1 - \gamma) (r + \sigma) \kappa + \gamma \kappa (r + \sigma + \lambda_w(\theta)) = (1 - \gamma) (p - b) \lambda_f(\theta)$$

$$(r + \sigma) \kappa + \gamma \kappa \lambda_w(\theta) = (1 - \gamma) (p - b) \lambda_f(\theta)$$

$$\frac{r + \sigma + \gamma \lambda_w(\theta)}{\lambda_f(\theta)} \kappa = (1 - \gamma) (p - b)$$

which is exactly the job creation condition. To explain this:

$$\underbrace{\frac{r + \sigma + \gamma \lambda_w(\theta)}{\lambda_f(\theta)}}_{\text{annuitized job creation cost}} \kappa = \underbrace{(1 - \gamma)(p - b)}_{\text{firm's share of match surplus}}$$

2.4 Transition Dynamics

Now let's consider the transition dynamics by not imposing ss. conditions. Collect:

$$\begin{aligned} rW(t) &= w(t) + \sigma(U(t) - W(t)) + W'(t) \\ rU(t) &= b + \lambda_w(\theta(t))(W(t) - U(t)) + U'(t) \\ rJ(t) &= p(t) - w(t) + \sigma(V(t) - J(t)) + J'(t) \\ rV(t) &= -\kappa + \lambda_f(\theta(t))(J(t) - V(t)) + V'(t) \\ (1 - \gamma)(\tilde{W}(w, t) - U(t)) &= \gamma(\tilde{J}(w, t) - V(t)) \\ V(t) &= 0 \end{aligned}$$

so

$$\begin{aligned} 0 &= -\kappa + \lambda_f(\theta(t))(J(t) - 0) \\ J(t) &= \frac{\kappa}{\lambda_f(\theta(t))} \end{aligned}$$

and

$$\begin{aligned} J'(t) &= rJ(t) - [p(t) - w(t) + \sigma(V(t) - J(t))] \\ &= (r + \sigma)J(t) - [p(t) - w(t)] \\ &= (r + \sigma) \frac{\kappa}{\lambda_f(\theta(t))} - p(t) + w(t) \end{aligned}$$

also given $(1 - \gamma)(W(t) - U(t)) = \gamma(J(t) - V(t))$ holds, we have that

$$\begin{aligned} (1 - \gamma)(W'(t) - U'(t)) &= \gamma(J'(t) - V'(t)) \\ (1 - \gamma)(W'(t) - U'(t)) &= \gamma J'(t) \end{aligned}$$

and

$$\begin{aligned} W'(t) - U'(t) &= rW(t) - [w(t) + \sigma(U(t) - W(t))] - rU(t) + b + \lambda_w(\theta(t))(W(t) - U(t)) \\ &= (r + \lambda_w(\theta(t)) + \sigma)(W(t) - U(t)) - w(t) + b \end{aligned}$$

so

$$\begin{aligned} \frac{\gamma J'(t)}{1-\gamma} &= (r + \lambda_w(\theta(t)) + \sigma) \frac{\gamma}{1-\gamma} \frac{\kappa}{\lambda_f(\theta(t))} - w(t) + b \\ \frac{\gamma J'(t)}{1-\gamma} + J'(t) &= \left[(r + \lambda_w(\theta(t)) + \sigma) \frac{\gamma}{1-\gamma} \frac{\kappa}{\lambda_f(\theta(t))} - w(t) + b \right] + \left[(r + \sigma) \frac{\kappa}{\lambda_f(\theta(t))} - p(t) + w(t) \right] \\ \frac{J'(t)}{1-\gamma} &= \frac{r + \sigma + \gamma \lambda_w(\theta(t))}{1-\gamma} \frac{\kappa}{\lambda_f(\theta(t))} + b - p(t) \\ J'(t) &= \frac{r + \sigma + \gamma \lambda_w(\theta(t))}{\lambda_f(\theta(t))} \kappa - (1-\gamma) [p(t) - b] \end{aligned}$$

I've been checking this and it seems that Toshihiko's note is having the wrong sign here. Anyway let's move on. Given $J(t) = \frac{\kappa}{\lambda_f(\theta(t))}$ we have

$$\begin{aligned} J(t) &= \frac{\kappa}{\lambda_f(\theta(t))} \\ J'(t) &= \frac{-\lambda'_f(\theta(t))\theta'(t)}{[\lambda_f(\theta(t))]^2} \\ &= \frac{\kappa}{\lambda_f(\theta(t))} \underbrace{\left(-\frac{\lambda'_f(\theta(t))\theta(t)}{\lambda_f(\theta(t))} \right)}_{\equiv \eta(\theta(t))} \frac{\theta'(t)}{\theta(t)} \\ &= J(t) \eta(\theta(t)) \frac{\theta'(t)}{\theta(t)} \end{aligned}$$

where $\eta(\theta(t)) = -\frac{\lambda'_f(\theta(t))\theta(t)}{\lambda_f(\theta(t))} > 0$ is the elasticity of matching function. So if $\theta(t) > \theta$ where θ is the ss. value of the tightness, we have $J'(t) > 0$ so that $\theta'(t) > 0$, similarly if $\theta(t) < \theta$, we have $\theta'(t) < 0$. So that the only case where $\theta(t)$ does converge is that $\theta(t) \equiv \theta$, then it's easy to see that $J'(t) = 0$ and $J(t)$ is fixed too.

Therefore, we can say that job-creation condition holds even not in ss. So given

$$\begin{aligned}\theta &= \frac{v(t)}{u(t)} \\ \frac{r + \sigma + \gamma \lambda_w(\theta)}{\lambda_f(\theta)} \kappa &= (1 - \gamma)(p - b) \\ u'(t) &= \sigma(1 - u(t)) - \lambda_w(\theta(t))u(t)\end{aligned}$$

one can pin down the dynamics of $(v(t), u(t), \theta)$. Now let's consider the case where $u(t) > u$, where u is given by the Beveridge curve condition:

$$u = \frac{\sigma}{\sigma + \lambda_w\left(\frac{v}{u}\right)}$$

then we have $u'(t) < 0$, similarly we have $u'(t) > 0$ when $u(t) < u$, so it's easy to establish that $u(t) \rightarrow u$ monotonically. Given $v = \theta u$, the dynamic of $v'(t)$ is similar.

3 Efficiency and Hosios condition

Notation. I use the notation of the model in my paper: worker bargaining power α , matching elasticity η , exogenous separation s , vacancy cost κ_v , and the Cobb–Douglas matching function $M(u, v) = \Psi u^\eta v^{1-\eta}$. Hence the result reads $\alpha = \eta$; in the baseline-notes notation of Notes 1–3 the same statement is $\gamma = \eta$, with γ the worker's Nash share. Write $p(\theta) = \theta q(\theta)$ for the job-finding rate, $q(\theta)$ for the vacancy-filling rate, and

$$\eta(\theta) \equiv -\frac{q'(\theta)\theta}{q(\theta)} \in (0, 1)$$

for the matching elasticity. The derivation is in continuous time, as in Notes 1–3. For a homogeneous benchmark I write y for the flow output of a match and b for the flow value of unemployment.

3.1 Object

From the steady-state *job-creation condition* under Nash bargaining and free entry was derived. Translating to model notation ($\gamma \rightarrow \alpha$, $\sigma \rightarrow s$, $\lambda_w \rightarrow p$, $\lambda_f \rightarrow q$, $\kappa \rightarrow \kappa_v$, productivity $\rightarrow y$, benefit $\rightarrow b$):

$$\frac{\kappa_v}{q(\theta)} = \frac{(1 - \alpha)(y - b)}{r + s + \alpha p(\theta)}. \quad (5)$$

This pins down the decentralized tightness θ^{dec} . The goal of this note is to derive the tightness a constrained planner would choose, θ^* , and compare the two.

3.2 Why efficiency can fail

The model so far is purely positive. Before asking *how far* θ^{dec} sits from θ^* , note *why* they differ at all. When one firm posts an extra vacancy it raises $\theta = v/u$, producing two uncompensated spillovers of opposite sign:

1. **Thick-market (trading) externality:** positive, on the *other* side. Higher θ raises $p(\theta) = \theta q(\theta)$: unemployed workers match faster. The firm is not paid for this.
2. **Congestion externality:** negative, on its *own* side. Higher θ lowers $q(\theta)$: every other firm fills more slowly. The firm bears none of this.

A third wedge comes from *pricing*: the firm sinks κ_v before matching, then Nash bargaining hands the worker share α of the surplus, so the firm recovers only $(1 - \alpha)$ of the value it created (**hold-up**).

3.3 The constrained planner's problem

The planner faces the **same** matching function (it cannot hand-match) and chooses the path of vacancies to maximize discounted net output.

Employed workers produce y , the unemployed produce b at home, and each vacancy costs κ_v .

$$\max_{\{\theta_t\}_{t=0}^{\infty}} \int_0^{\infty} e^{-rt} \left[y(1 - u_t) + b u_t - \kappa_v \theta_t u_t \right] dt$$

subject to:

$$u'_t = \underbrace{s(1 - u_t)}_{\text{inflow}} - \underbrace{\theta_t q(\theta_t) u_t}_{\text{outflow}}.$$

State $x = u$, control $c = \theta$.

Form the current-value Hamiltonian with costate ξ on u :

$$\mathcal{H} = y(1 - u) + bu - \kappa_v \theta u + \xi [s(1 - u) - \theta q(\theta) u].$$

Since more unemployment is bad, $\xi < 0$; define the social value of an additional *employed* worker $\lambda \equiv -\xi > 0$.

By Pontryagin's maximum principle, we have the following:

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial \theta} &= -\kappa_v u - \xi u \frac{\partial}{\partial \theta} [\theta q(\theta)] = 0 \\ \xi' &= r\xi - \frac{\partial \mathcal{H}}{\partial u} \\ u' &= s(1 - u) - \theta q(\theta) u\end{aligned}$$

Then by doing little math:

$$\begin{aligned}\frac{\partial}{\partial \theta} [\theta q(\theta)] &= q(\theta) + \theta q'(\theta) = q(\theta) \left[1 + \frac{\theta q'(\theta)}{q(\theta)} \right] = q(\theta) [1 - \eta(\theta)] \\ \frac{\partial \mathcal{H}}{\partial u} &= -y + b - \kappa_v \theta - \xi s - \xi \theta q(\theta)\end{aligned}$$

so

$$\begin{aligned}-\kappa_v u &= \xi u \frac{\partial}{\partial \theta} [\theta q(\theta)] \\ &= \xi u q(\theta) [1 - \eta(\theta)] \\ -\kappa_v &= \xi q(\theta) [1 - \eta(\theta)] \\ \frac{\kappa_v}{q(\theta)} &= \lambda (1 - \eta(\theta))\end{aligned}$$

and

$$\begin{aligned}\xi' &= r\xi - [-y + b - \kappa_v \theta - \xi s - \xi \theta q(\theta)] \\ &= \xi (r + s + \theta q(\theta)) + y - b + \kappa_v \theta\end{aligned}$$

at ss. we have $\xi' = 0$ so

$$\begin{aligned}
0 &= \xi(r + s + \theta q(\theta)) + y - b + \kappa_v \theta \\
-\xi(r + s + \theta q(\theta)) - \kappa_v \theta &= y - b \\
\lambda(r + s + \theta q(\theta)) - \lambda(1 - \eta(\theta))\theta q(\theta) &= y - b \\
\lambda[(r + s + \theta q(\theta)) - (1 - \eta(\theta))\theta q(\theta)] &= y - b \\
\lambda[r + s + \eta(\theta)\theta q(\theta)] &= y - b
\end{aligned}$$

so

$$\begin{aligned}
\frac{\kappa_v}{q(\theta)} &= \lambda(1 - \eta(\theta)) \\
&= \left[\frac{y - b}{r + s + \eta(\theta)\theta q(\theta)} \right] (1 - \eta(\theta)) \\
&= \frac{(y - b)(1 - \eta(\theta))}{r + s + \eta(\theta)\theta q(\theta)}
\end{aligned}$$

3.4 Comparison: the Hosios condition

Place the two job-creation conditions side by side:

$$\begin{aligned}
\frac{\kappa_v}{q(\theta)} &= \frac{(1 - \alpha)(y - b)}{r + s + \alpha p(\theta)} \\
\frac{\kappa_v}{q(\theta)} &= \frac{(y - b)(1 - \eta(\theta))}{r + s + \eta(\theta)\theta q(\theta)}
\end{aligned}$$

They are **identical except that the decentralized α is replaced by the elasticity η** . The two coincide for all parameter values if and only if

$$\boxed{\alpha = \eta}$$

and this is the ***Hosios condition***.

Interpretation. Each side of the market should capture a surplus share equal to its marginal contribution to producing matches. The worker's search input enters $M(u, v)$ with elasticity η ; paying the worker $\alpha = \eta$ makes her internalize exactly the externality she imposes, and the congestion and thick-market effects cancel.

3.5 Signing the distortion

Then the direction of the inefficiency when $\alpha \neq \eta$ must be *proved*. Define, for a standing for either α (market) or η (planner),

$$\Phi(\theta, a) \equiv \frac{\kappa_v}{q(\theta)} - \frac{(1-a)(y-b)}{r+s+ap(\theta)}$$

where we set $\Phi(\theta, a) = 0$. Because the matching function is Cobb–Douglas, $M = \Psi u^\eta v^{1-\eta}$, the elasticity $\eta(\theta) \equiv -\theta q' / q = \eta$ is constant. Hence η is a θ -independent parameter, the planner's condition is $\Phi(\theta, \eta) = 0$ with η fixed, and the comparison " $\alpha = \eta$ " together with the IFT below hold globally. Then

$$\frac{\partial \Phi}{\partial \theta} = -\frac{\kappa_v q'}{q^2} + \frac{(1-a)(y-b)ap'}{(r+s+ap)^2}$$

given $q' < 0, p' > 0, y > b$, we have

$$\frac{\partial \Phi}{\partial \theta} > 0$$

and also

$$\begin{aligned} \frac{\partial \Phi}{\partial a} &= \frac{(y-b)(r+s+ap) + (1-a)(y-b)p}{(r+s+ap)^2} \\ &= \frac{(y-b)(r+s+ap+p-ap)}{(r+s+ap)^2} \\ &= \frac{(y-b)(r+s+p)}{(r+s+ap)^2} > 0 \end{aligned}$$

so that by implicit function theorem,

$$\frac{d\theta}{da} = -\frac{\partial \Phi}{\partial a} / \frac{\partial \Phi}{\partial \theta} < 0$$

so the solution θ is strictly decreasing in a , since the market uses $a = \alpha$ and the planner uses $a = \eta$, then $\alpha > \eta$ means $\theta^{dec} < \theta^*$, which means that there are too few vacancies and vice versa.